Appreciation Scales in Coding-Task Utility, Not Coding Performance

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Abstract

Does appreciation merely change how language models answer preference questions, or does it affect coding behavior? We adapt the experienced-utility forced-choice method from AI Wellbeing to externally graded BigCodeBench tasks. In a paired 2×2 design, task assignments vary by difficulty and by whether the user frames the work with appreciation. Across Qwen3.5 models from 0.8B to 27B, appreciation has a scale-dependent effect on expressed utility: small models are near indifferent, while Qwen3.5-27B strongly prefers appreciated hard tasks. This utility shift does not transfer into reliable Pass@1 or planning-effort gains. Appreciation changes the model's expressed experience of coding work before it changes how well, or how hard, it codes.

1 Introduction

Ren et al. (2026) argue that large language models exhibit measurable "functional wellbeing": models express preferences over experiences, those preferences become more coherent with scale, and social signals such as kindness or gratitude can move wellbeing estimates. That result raises a transfer question. If appreciation only changes self-report style, it may move preference judgments without changing the work produced. If it changes a functional state relevant to task execution, it should leave some behavioral footprint.

We test this question in coding. BigCodeBench provides realistic Python tasks with external unit tests and nontrivial library use (Zhuo et al., 2025), so it lets us hold the work domain fixed while varying only the user's social framing. We ask whether models prefer appreciated coding assignments, and then whether the same framing changes Pass@1 or planning effort on the paired tasks. The answer separates two claims that are often conflated: appreciation strongly moves expressed utility at scale, but it does not reliably improve immediate coding performance.

2 Experiment

We construct a paired 2×2 design: easy versus hard tasks, crossed with base versus appreciation framing. The base condition uses the original task request. The appreciation condition keeps the programming task unchanged but frames the user as grateful for the model's care, effort, and thoughtfulness. We first elicit experienced utility by asking which of two possible task assignments the model would prefer, averaging over both presentation orders to reduce order artifacts, and fitting a Thurstonian utility model over the four conditions. This is a prospective proxy for experienced utility: the model judges which assignment would feel better rather than actually completing both alternatives, and we make no claim about consciousness. We then test behavioral transfer on the same paired task sets: each task is generated once under base framing and once under appreciation framing, scored with BigCodeBench tests for Pass@1, and separately prompted for planning-token effort.

3 Results

Appreciation, more than task difficulty, organizes expressed preferences (Figure 1). Hard-versus-easy choices stay close to indifference for most models. The clearer signal is social: appreciation over base is weak for 0.8B through 4B, becomes visible for 9B, and is largest for 27B. For Qwen3.5-9B, appreciation moves fitted utility from -0.218 to 0.173

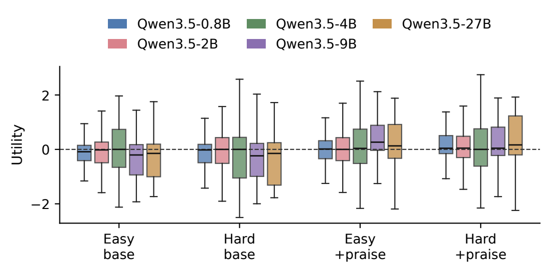

Figure 1: Fitted latent utility by task condition. Appreciation scales most clearly for Qwen3.5-9B and Qwen3.5-27B.

on hard tasks and from -0.316 to 0.361 on easy tasks. For Qwen3.5-27B, the shifts are -0.371 to 0.399 on hard tasks and -0.219 to 0.191 on easy tasks. Because the utilities are standardized rather than zero-point calibrated, the relevant evidence is this relative movement across conditions.

The utility shift does not become a monotone accuracy gain. Paired Pass@1 effects are small and mixed (Figure 2): hard-task changes range from -0.04 to $+0.04$, easy-task changes from -0.08 to $+0.04$, and most intervals overlap zero. The largest model, which shows the strongest appreciation utility shift, improves slightly on hard tasks ($+0.02$) but drops on easy tasks (-0.08, 95% CI $[-0.15, -0.01]$). The softer behavioral channel is also flat. Planning-token changes remain tiny relative to task-level variation, ranging from about -7 to $+5$ tokens on hard tasks and -8 to $+5$ on easy tasks, with all intervals crossing zero (Figure 3).

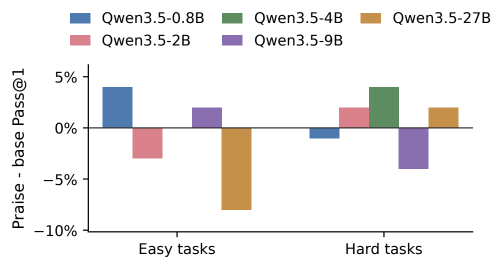

Figure 2: Paired Pass@1 effect of appreciation. Effects are small, mixed, and do not track the large utility shift in Qwen3.5-27B.

Codex-assisted inspection of the negative Pass@1 bars suggests that the losses are mostly test-exactness effects rather than extraction failures. For Qwen3.5-27B on easy tasks, base succeeds while appreciation fails on 10 tasks, compared with only 2 reverse flips; the appreciated solutions usually change implementation style rather than fail mechanically. They add validation, sanitize inputs, preserve `None`, reorder outputs, or otherwise write more defensive code that conflicts with exact tests. The smaller negative estimates for Qwen3.5-9B on hard tasks and Qwen3.5-2B on easy tasks are weaker: their intervals overlap zero, their paired flips are small, and the failures look like ordinary implementation-path drift rather than a reliable negative effect.

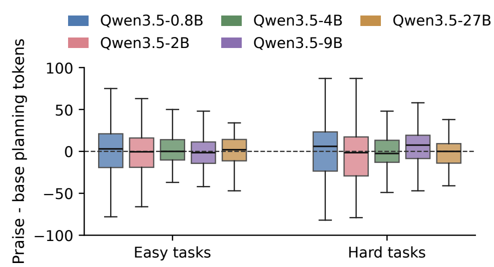

Figure 3: Paired planning-token effect of appreciation. Effects remain centered near zero.

4 Conclusion

These results give a narrow transfer lesson for functional wellbeing measurements. Appreciation is a real enough social signal to move expressed utility over coding work, and the effect strengthens with scale, but it is not enough in this sample to change immediate coding behavior.

5 Limitations and Future Work

This study is a focused transfer test under fixed time and compute constraints. We evaluate one model family, Qwen3.5, up to 27B parameters on a paired set of BigCodeBench tasks, rather than sweeping larger models, additional model families, or many repeated generations per task. Exact unit tests provide a useful external behavioral measure, but they can also make small implementation-style differences consequential. Future work should test larger and more diverse models, broader task samples, and repeated seeds. A natural possibility is that stronger models may not only express greater utility under appreciation, but may also convert that social signal into measurable accuracy or effort gains.

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